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Development and Evaluation of an Affordable VR Glove with Force Feedback and IMU Tracking

Master's Thesis

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ChatGPT v1.0	Writing Equations in LaTeX	Pasted equations from references and asked ChatGPT to convert them into LaTeX format.
ChatGPT v1.0	Spelling and Spacing Check	Drafted the conclusion and used ChatGPT to fix spacing and typing mistakes in LaTeX syntax.

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1 Abstract

Advancements in Virtual Reality (VR) technology have revolutionized medical training, particularly in neurosurgery, by providing immersive, interactive, and risk-free environments for skill development. However, the widespread adoption of VR-based training tools, especially haptic gloves with precise hand-tracking and force-feedback mechanisms, is limited by their high cost, and provide limited force feedback, with most relying only on vibrotactile sensations. Additionally, many are not wireless, have higher interaction latency, and are not designed for easy assembly, making them less accessible for widespread use.

To address these challenges, a new approach was developed to create a haptic glove inspired by open-source initiatives like Lucid VR, which focus on affordable, DIY haptic VR gloves for hand tracking and force feedback. The original design relied on external hardware, such as the Valve Index controller or a Vive tracker, for hand rotation and position detection. This project aimed to replace those external trackers with Inertial Measurement Units (IMUs) for hand orientation and position tracking. The proposed system integrates the ICM20948 (9 DoF) and MPU6050 (6 DoF) IMUs, enabling a comparison of their performance in terms of orientation and position detection. Additionally, integrating a logic algorithm to mitigate common IMU-related issues such as drift and gimbal lock, ensuring more accurate tracking.

Finally, the proposed logic was evaluated by comparing it to the Vive tracker as a reference. The evaluation involved comparing traditional methods of calculating orientation using complementary filters and Kalman filters in terms of mean absolute error (MAE) error and latency from the reference. This comparison aimed to assess how well the proposed logic performs relative to commercial hardware, ensuring that the new system meets the necessary standards for effective VR training in medical applications.

2 Introduction

2.1 VR in Neurosurgical training

2.1.1 Neurosurgery challenges

Neurosurgery presents significant challenges due to the structural complexity of the brain and spinal cord. Precision is critical, as even minor mistakes can have severe consequences. Traditional methods, such as cadaver dissection, fall short of replicating live surgery conditions, particularly in terms of tissue elasticity and bleeding. Furthermore, limited mentorship, restricted access to cadaver labs, and inconsistent patient availability contribute to the inconsistencies in surgical training. Despite these limitations, many programs still adhere to the traditional apprenticeship model, aiming for mastery through prolonged hands-on experience and often summarized by the widely recognized goal of 10,000 hours of practice [1]. In addition to these challenges, there is a severe shortage of neurosurgeons, particularly in low- and middle-income countries (LMICs), where the ratio can be as low as one neurosurgeon per one million people [2]. This shortage significantly exacerbates the difficulty of providing adequate and consistent surgical training.

Virtual Reality (VR) offers a promising solution to many of these issues by creating an immersive, risk-free environment where trainees can repeatedly practice complex neurosurgical procedures. VR simulations not only reduce the reliance on cadavers or patients for training but also offer structured feedback through the quantitative assessment of surgical performance [3]. A meta-analysis of 14 studies has demonstrated that simulation-based training proved to be more effective than traditional apprenticeship methods in developing critical surgical skills [4].

2.1.2 Brief History

Virtual Reality (VR) has a long history, with its first application dating back to 1929 as a flight simulator. The term "virtual reality" was formally introduced in 1987 [5], marking the start of its broader development, led by Tom Furness, a pioneering electrical engineer from the U.S. Air Force. Often referred to as the "Godfather of Virtual Reality" [6], Furness laid the foundation for VR's future applications.

In medicine, VR made its debut in 1980, thanks to orthopedic specialist Robert Mann, who integrated VR with a head-mounted device for medical use [7]. Over time, its application in healthcare expanded, including its use in the treatment of phobias, such as arachnophobia, in 1998 [8]. This gradual adoption of VR in both training and treatment settings has paved the way for its current use in advanced medical education, such as neurosurgical training.

2.1.3 Virtual Reality in Neurosurgery Training

Virtual reality (VR) made its first significant impact on neurosurgical disorder treatment in 2009, when David B. Clarke utilized the NeuroTouch neurosurgical simulator to treat a patient with a left frontal meningioma, a type of brain tumor that forms in the meninges, the protective layers surrounding the brain and spinal cord [9]. Since then, VR has become a valuable tool in neurosurgery, offering the ability to quantify surgeon performance, assess surgical proficiency, and track training progression [10, 11]. One study explored VR training with seven operating room (OR) nurses, who had an average of three years of experience in complex neurosurgical procedures, such as craniotomies. A craniotomy is a surgical procedure where part of the skull is temporarily removed to access the brain. The study demonstrated that VR significantly improved the nurses' confidence, particularly in areas such as recognizing intra-operative bleeding (from 42.9% pre-VR to 71.5 % post-VR) and assisting as scrub nurses during craniotomies (from 57.2 pre-VR to 85.7 % post-VR) [12]. However, knowledge related to setting up a craniotomy and identifying hemostatic agents (substances used to stop bleeding) remained unchanged.

For novice neurosurgeons, VR has been shown to reduce cognitive load (the mental effort required to perform tasks) and alleviate operative stress, enhancing their learning efficiency and confidence during complex procedures [13]. Additionally, VR facilitates global connectivity through tele-proctoring, which enables surgeons to receive real-time guidance on complex neurosurgical procedures, regardless of their geographical location. This is especially beneficial for surgeons in low- and middle-income countries [14]. One of the most significant advantages of VR is its capacity to simulate an unlimited number of cases, allowing surgeons to practice rare and complex procedures repeatedly, without the restrictions of physical models or real patients [15]. Research also indicates that younger surgeons show greater enthusiasm for learning through augmented reality (AR) systems, particularly those that utilize head-mounted displays (HMDs) for visualizations, making this a promising approach for engaging the next generation of medical professionals [16]. Furthermore, VR has shown potential in aiding "awake craniotomies," a procedure where patients remain conscious during surgery, allowing surgeons to map and preserve critical cognitive functions, such as language. Studies like those by Katsevman et al. demonstrated that VR can enhance cognitive screening during brain tumor surgeries, improving intraoperative decision-making by allowing surgeons to test functions such as language in real-time [17].

Lastly, a 2019 study by Fiederer et al. highlighted the potential of VR in neurosurgery by constructing detailed 3D models from MRI and CT data, which two neurosurgeons found to be highly accurate and immersive. These 3D models allow for a better understanding of complex anatomical structures and can be easily translated into 2D images for use during surgery. Their framework, built using open-source tools, emphasizes the power of VR for enhanced visualization in neurosurgical practice [18].

2.1.4 State of the Art for Neurosurgical training using VR

Several research groups have made significant strides in using virtual reality (VR) for neurosurgery, particularly in developing highly realistic 3D visualizations of brain structures [19, 20, 21]. Among these advancements, B. K. Hendricks' models utilize cutting-edge computer graphics and digital modeling technologies to create anatomically precise representations of cranial bones (the bones that form the skull) and cerebrovascular structures (the network of blood vessels supplying the brain) [22, 23].

In addition to research-based frameworks, a variety of commercial VR platforms have emerged for neurosurgical applications. Notable platforms include Dextroscope® [24, 25, 26, 27, 28], a tool used for intervention simulations, enabling surgeons to practice procedures before performing them on real patients. Another key platform, NeuroTouch, is designed for craniotomy-based procedures and tumor resections (the surgical removal of a tumor), incorporating haptic feedback, which provides tactile sensations to simulate the feeling of manipulating tissues and surgical instruments [29, 30, 31]. Similarly, ImmersiveTouch® combines augmented reality (AR) and immersive VR environments with haptic feedback, making it highly suitable for neurosurgical training [32, 33].

Other platforms, such as NeuroVR, Surgical Theater, Osso VR, and Touch Surgery, have also gained recognition in neurosurgical training. However, many of these platforms lack extensive scientific validation, meaning they have not been widely tested in research studies to confirm their effectiveness or accuracy in surgical education.

2.2 Haptic feedback and hand tracking in VR

The integration of haptic feedback and precise hand tracking is essential for enhancing the realism of VR simulations, particularly in fields like surgical training, where precision is paramount. Studies have demonstrated the importance of quantifying surgical skills and highlight the ongoing need for improved simulators that deliver realistic haptic feedback [34, 35, 36, 37]. The absence of such feedback in many VR surgical simulators diminishes immersion and reduces the accuracy of performance assessments in training environments [38].

When developing haptic gloves for surgical training, understanding the anatomical structure of the human hand is crucial. Each hand has 21 degrees of freedom, including the wrist, with each finger containing three bones distal, middle, and proximal phalanges and three joints, while the thumb has two bones and two joints [39]. This anatomical complexity, combined with the kinematic behavior of the hand, must be considered when designing effective force-feedback gloves. The movement range and accuracy of finger joints, particularly at the metacarpophalangeal (MCP), proximal interphalangeal (PIP), and distal interphalangeal (DIP) joints, define the specifications required for tracking systems.

These quantified movement ranges and joint precision are vital for developing gloves that can accurately replicate complex hand movements. This is particularly critical in neurosurgical training, where the ability to feel and respond to minute changes in tissue and anatomy can greatly enhance the learning experience and surgical outcomes. Haptic feedback in VR not only improves skill acquisition but also ensures that trainees can practice with a level of realism that closely mirrors actual surgical procedures, ultimately leading to better preparedness and proficiency in the operating room.

2.2.1 Categorization of Haptic Gloves

Force feedback gloves can be categorized based on the location of their actuation systems, which directly influence their functionality, weight, and ability to deliver precise feedback [40]. The main categories include:

1. **Ground-based systems:** These include gloves such as the HIRO III, where the glove is mounted on a fixed base. They offer strong force feedback but have limited mobility [41].
2. **Dorsal-based systems:** Examples include CyberGrasp and Sensor Glove II, which place actuation on the back of the hand. This enhances portability and wearability but often limits their ability to simulate multi-dimensional forces [42, 43].
3. **Palm-based systems:** Such as the Rutgers Master II, these systems ground the actuation in the palm to simulate opposition grips. However, they have restricted motion [44].
4. **Digit-based systems:** Like DESR, these gloves provide force feedback between individual fingers, allowing for more precise tactile interaction [45].

In addition to actuation positioning, force feedback systems utilize various sensing techniques to track hand and finger movements with high precision. For example, one study demonstrated the use of rotary position sensors placed at the metacarpophalangeal (MCP) joints to detect flexion, extension, adduction, and abduction movements by measuring changes in resistance as the joints move. To improve accuracy and account for differences in finger lengths, a Min-Max Scaling (MMS) filter was applied to scale sensor data between 0 and 1, enabling precise motion tracking in virtual reality environments. The system achieved a mean absolute error (MAE) of 3.091° for flexion/extension and 2.068° for adduction/abduction, with a processing delay of 145.37 µs per finger and an overall hand tracking latency of 4 ms [46]. Another design uses Leap Motion's optical tracking to capture hand movements, combined with McKibben muscles for kinesthetic feedback, which allows users to feel resistance when interacting with virtual objects. Testing with 15 users showed a significant enhancement in immersion, particularly during tasks like pressing virtual buttons, with feedback described as "mesmerizing" and "amazing," although slight delays were noted by some users [47].

Additionally, some systems employ MR (magnetorheological) brakes along with CyberGlove for tracking finger movements and providing force feedback in VR. The MR brakes create adjustable resistance to simulate interactions with virtual objects, while the CyberGlove tracks finger positions, reducing task completion times by 79 % and enabling users to distinguish object stiffness with 80 % accuracy [48]. Another wearable module uses fiber Bragg grating (FBG) strain sensors for accurate finger joint tracking, employing algorithms like Point Strain (PTS) and Proportional Summation (PS) to enhance accuracy.

This system achieves high precision, with an average error of 0.47 ± 2.51 degrees and a mean absolute error (MAE) of 1.63 ± 1.97 degrees [49]. Lastly, the SenseGlove NOVA, combined with VIVE trackers, offers a VR-based neurosurgical training simulator. These gloves provide vibrotactile feedback and track finger movements, while VIVE trackers capture hand position and orientation. The system adjusts resistance based on finger movement and object properties, simulating interactions with surgical tools like drills and catheters.Although the gloves were found to enhance realism, some users mentioned limitations such as the lack of hand size adjustments [50].

2.2.2 State of the art Haptic feedback gloves

Current state-of-the-art VR gloves incorporate a range of technologies aimed at enhancing immersion and interaction in virtual environments. Notable examples include GloveOne [50], CaptoGlove [51], Manus VR [52], Senso [53], Noitom Hi5 VR [54], VRfree [55], HaptX [56], and the Nova Sese Glove [57]. These gloves generally feature 9-DOF inertial measurement units (IMUs) and flex sensors to track finger movements and hand orientation, providing precise control and feedback in VR systems. Many, such as GloveOne, Senso, and Manus VR, incorporate vibration motors for tactile feedback, enhancing the sensation of interacting with virtual objects. CaptoGlove extends its functionality beyond VR by enabling control of smart devices, while VRfree focuses on wireless, low-latency tracking for seamless interaction. HaptX, at the high end, stands out by delivering realistic tactile feedback through microfluidic actuators, simulating sensations of pressure and touch. Meanwhile, Nova Sese Glove offers an accessible, easy-to-integrate solution for VR platforms, balancing tactile feedback and motion tracking.

2.2.3 Disadvantage of the existing commercial devices

- No built-in interpretation of human bio signals.
- Lack of feedback that allows users to perceive virtual objects as real.
- Limited use of force feedback (most rely on vibrotactile feedback, except for Dexmo).
- High cost, making them less accessible.
- Not designed for easy assembly from kits.
- Some gloves are not wireless, leading to higher latency in interactions.

2.3 Proposed System

The proposed system is a low-cost, open-source VR glove inspired by Lucas VR Tech's Lucid VR design [58]. Lucid VR gloves are well-known for their affordability and open-source nature, making advanced VR experiences accessible to a wider audience. These gloves track finger movements using potentiometers, strings, and 3D-printed parts, offering an effective yet inexpensive way to simulate real hand motions in virtual reality. However, the original design relies on HTC Vive trackers for hand movement and orientation tracking, which increases the system's complexity and cost.

To improve on this, the proposed system replaces these external trackers with Inertial Measurement Units (IMUs) for hand orientation and position tracking. Specifically, it integrates two different IMUs—the ICM20948 (9 DoF) and MPU6050 (6 DoF)—to conduct a comparative analysis of their performance in detecting hand movement and orientation. The 9 DoF IMU provides additional magnetometer data, which has the potential to enhance accuracy, while the 6 DoF IMU relies solely on accelerometer and gyroscope data. This comparison helps determine whether the additional sensor data significantly improves tracking or if a 6 DoF IMU is sufficient for accurate hand movement detection. Since IMUs commonly suffer from issues such as drift and gimbal lock, a custom logic algorithm was integrated to mitigate these challenges and improve tracking accuracy.

3 Hardware Overview

This chapter presents a detailed analysis of the hardware components used in constructing the VR gloves, covering key elements such as microcontrollers, sensors, servo motors, power sources, and communication modules. It also explores the custom 3D-printed components designed to house these elements, including the glove frame and sensor mounts. Additionally, the chapter outlines the assembly process, explaining the electrical connections established through soldering and wiring, along with the mechanical design considerations that ensure both comfort and flexibility.

3.1 Component Selection

The essential hardware components required for assembling the VR gloves are summarized in the table below. The system utilizes WL (Brass) potentiometers for finger tracking, 9g MG90S servo motors for haptic feedback, and an ESP32-WROOM-32 microcontroller for wireless connectivity with SteamVR via Open Gloves for VR visualization. For mechanical actuation, servo motors are combined with inexpensive retractable badge reels, which function as rotary springs. To enhance stability and comfort, Velcro straps and foam sheets are incorporated into the design. A 5V power supply is necessary to efficiently operate the servo motors. The following table provides a detailed breakdown of the hardware components, specifying their quantities, descriptions, and purchase links for easy sourcing.

Component	Quantity
3D Printing Materials: PLA Filament	1 kg Spool
9g Servo Motors (MG90S)	5 per hand
ESP32 WROOM 32 Development Board	1 per hand
WL (Brass) Potentiometers	5 per hand
Retractable Badge Reels	5 per hand
Power Supply	1 per hand
Sports Gloves	1 pair
WD-40	1
Velcro Fasteners	1
ICM20948 (IMU sensor)	1
MOU6050 (IMU sensor)	1

Tab. 3.1: List of Hardware Components for VR Gloves

All components were chosen based on Lucid VR's open-source framework, with modifications introduced to replace the Valve Index controller with an IMU sensor for position and rotation tracking. This adaptation required careful consideration in selecting an IMU sensor that offers the best balance between performance, cost, and accuracy. A detailed discussion on the selection criteria and performance evaluation of the IMU sensor is provided in the following chapter.

3.2 Mechanical Design and 3D-Printed Components

In addition to the electronic components, several 3D-printed parts were essential for assembling the VR gloves, offering structural support and facilitating mechanical movement. These custom-printed components ensure the secure placement of sensors, servos, and other hardware while maintaining flexibility and comfort. The table below lists the names and quantities of the 3D-printed parts used in the assembly.

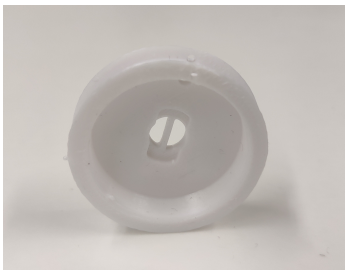
3D Part Name	Quantity Printed
Guide Nodes	15 per hand
Tensioners	5 per hand
Spools	5 per hand
Covers	5 per hand
Holders	5 per hand
Rigid Mounts	1 per hand
End caps	5 per hand

Tab. 3.2: List of Hardware Components for VR Gloves

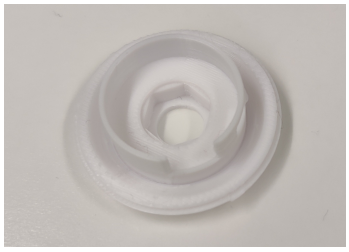
These 3D-printed components were fabricated using a Monoprice V1 3D printer with PLA filament, maintaining a nozzle temperature of 200°C and a bed temperature of 65°C. Initial prints had defects due to warping, which were resolved by adjusting print speed and cooling settings. Each part plays a critical role in ensuring precise tracking and haptic feedback. The guide nodes Figure 3.1 (a) direct the movement of the tensioning strings, preventing misalignment and ensuring smooth operation. The spools Figure 3.1 (b) wind and unwind the strings, while the tensioners Figure 3.1 (c) regulate string tightness, allowing for controlled force feedback. The covers Figure 3.1 (d) protect internal mechanisms from dust and damage, and the holders Figure 3.1 (e) secure key components in place, maintaining system stability. The rigid mount Figure 3.1 (f) serves as a sturdy base, anchoring the entire glove assembly, while the end caps Figure 3.1 (g) prevent the strings from slipping, ensuring consistent performance. By leveraging 3D printing, this design ensures cost-effectiveness, ease of customization, and rapid prototyping, making it a viable alternative to expensive commercial VR gloves.



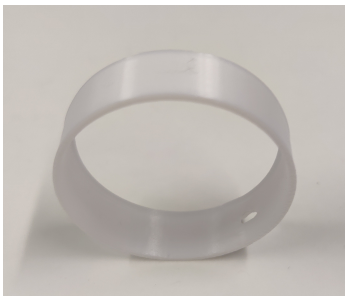
(a) 3D printed: Guide Node



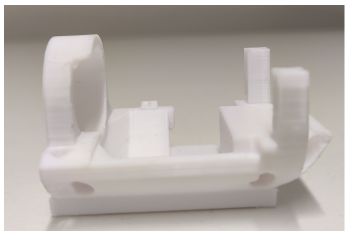
(b) 3D printed: Spool



(c) 3D printed: Tensioners



(d) 3D printed: Cover



(e) 3D printed: Holder



(f) 3D printed: Rigid Mount

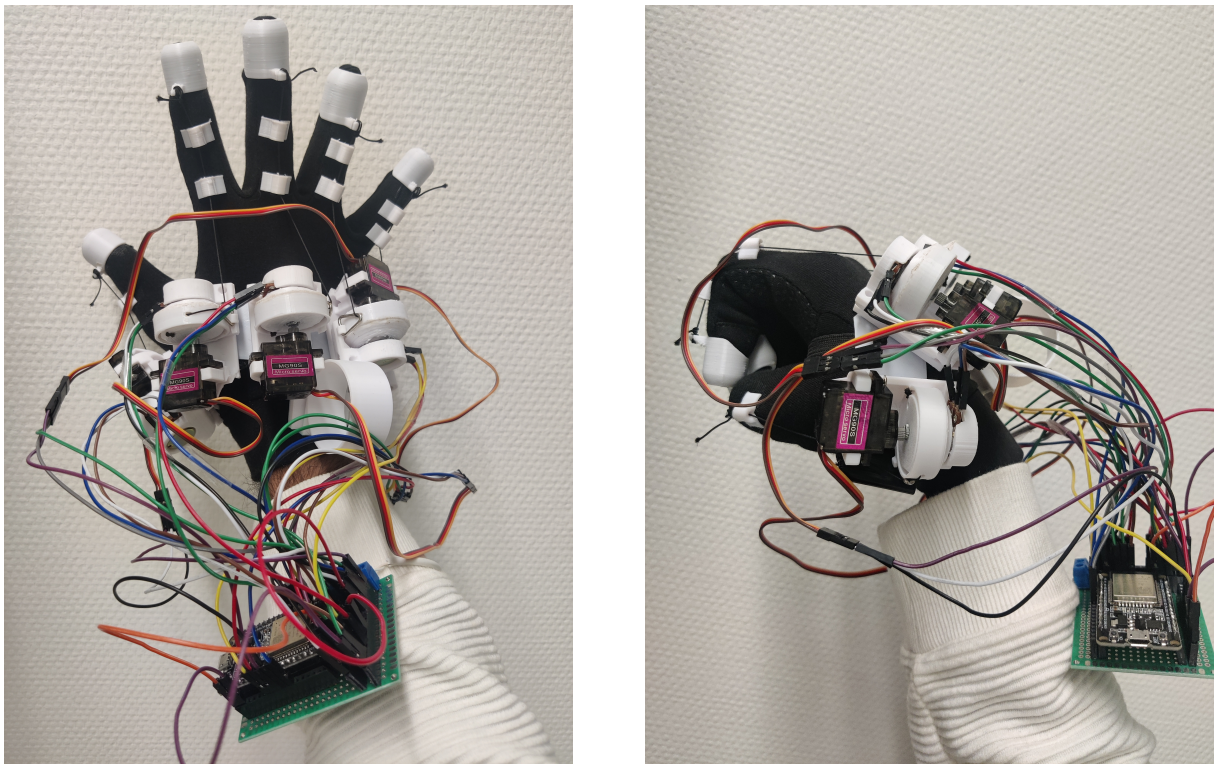


(g) 3D printed: End Cap

Fig. 3.1: 3D Printed parts

3.2.1 Assembly of 3D printed parts

For assembling the modules, the potentiometer should be fitted into the designated holder piece, using the round hole for WL potentiometers. The metal loops from the badge reel ends need to be removed using pliers and then clamped onto the plastic loops on the holders. The strings should be run through these metal loops to complete the setup. To set up the rigid mount, a layer of foam should be glued to the bottom for added comfort. Optionally, a loop of elastic can be inserted through the two circular holes to sit in front of the thumb. A Velcro strap should be attached between the two loops of the rigid mount and trimmed to ensure a snug fit around the hand. Once mounted, the modules can be slid onto the rigid mount, adjusting some backward if necessary to ensure proper string alignment.The nodes and caps must be glued properly to secure all components in place. Stringing the glove correctly is essential to achieving accurate tracking and haptic feedback. The haptic system should be installed only after flashing the firmware onto the glove. Once finger tracking is functional, servo motors should be fitted into their respective slots, using a single-direction servo head placed on the gear. Adjustments may be necessary to ensure proper movement, using the extend/retract servos fully function in the open Glove UI to calibrate the positioning. When fully extended, the servos should completely limit the hand from opening, and when fully retracted, there should be no limitation on hand movement. Following these steps ensures a proper assembly of the Haptic Glove, optimizing both tracking accuracy and haptic feedback performance. Figure 7.2 shows the fully assembled haptic glove with all modules in place.



(a) Front view of the glove

(b) Demonstration of flexion movements

Fig. 3.2: Complete assembly of the glove

3.2.2 Wiring connection between sensors and micro controller

The wiring of the haptic glove is crucial for its functionality, specifically for the potentiometers and servo motors. Each finger is equipped with a 10 kΩ potentiometer to measure finger bending, with one terminal connected to **GND**, the other to **3.3V**, and the wiper (middle pin) connected to an **ADC (Analog-to-Digital Converter)** pin on the ESP32. The connections are as follows:

- **RV_Pinky** → GPIO36 (ADC0)
- **RV_Ring** → GPIO39 (ADC3)
- **RV_Middle** → GPIO34 (ADC6)
- **RV_Pointer** → GPIO35 (ADC7)
- **RV_Thumb** → GPIO32 (ADC4)

These connections allow the ESP32 to read voltage variations as the user moves their fingers, enabling accurate tracking.

The servo motors, responsible for haptic feedback, are powered through **Powerbank_2** at **+5V**, with their ground wires connected to **GND**. The PWM (Pulse Width Modulation) signal wires are connected to specific GPIO pins:

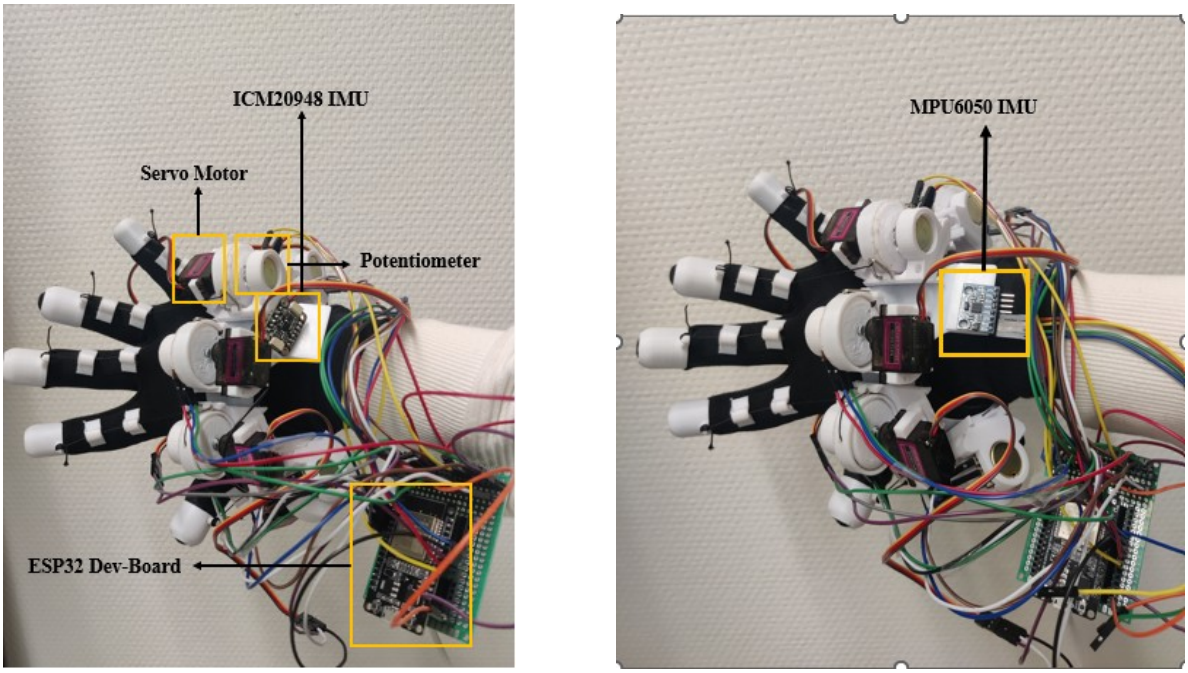
- **Servo_Pinky** → GPIO13
- **Servo_Ring** → GPIO12
- **Servo_Middle** → GPIO14
- **Servo_Pointer** → GPIO27
- **Servo_Thumb** → GPIO26

The PWM signals from the ESP32 control the position of each servo, restricting or allowing movement based on user interactions. This wiring configuration ensures seamless communication between the components, enabling effective finger tracking and responsive haptic feedback for an immersive VR experience.

To accurately track hand orientation, an Inertial Measurement Unit (IMU) sensor is integrated into the glove. The IMU communicates with the ESP32 via the I2C (Inter-Integrated Circuit) protocol, using the following connections:

- **IMU SDA** → GPIO21 (I2C Data Line)
- **IMU SCL** → GPIO22 (I2C Clock Line)
- **IMU VCC** → 3.3V
- **IMU GND** → GND

The IMU continuously sends accelerometer, gyroscope and magnetometer data to the ESP32, which processes the signals to determine hand orientation. The combination of finger tracking and hand motion sensing enhances the realism and functionality of the VR glove. Figure 3.3 shows the complete wired connections.



(a) IMU ICM20948 attached to the glove **(b)** IMU MPU6050 attached to the glove

Fig. 3.3: Complete figure after the wiring

4 IMU Tracking

4.1 What is an IMU?

An Inertial Measurement Unit (IMU) is a device that measures angular velocity and linear acceleration along three axes. It may also include a magnetometer to measure Earth's magnetic field for improved orientation tracking. An IMU typically consists of:

- **A three-axis gyroscope** – measures angular velocity.
- **A three-axis accelerometer** – measures linear acceleration.
- **A temperature sensor** – compensates for sensor drift.
- **Optionally, a three-axis magnetometer** – measures Earth's magnetic field and corrects orientation drift, acting as a digital compass.

IMUs provide high-accuracy orientation tracking by capturing both linear acceleration and rotational movements. They are widely used to determine and control the attitude and trajectory of moving bodies in applications like robotics, virtual reality, drones, and navigation systems.

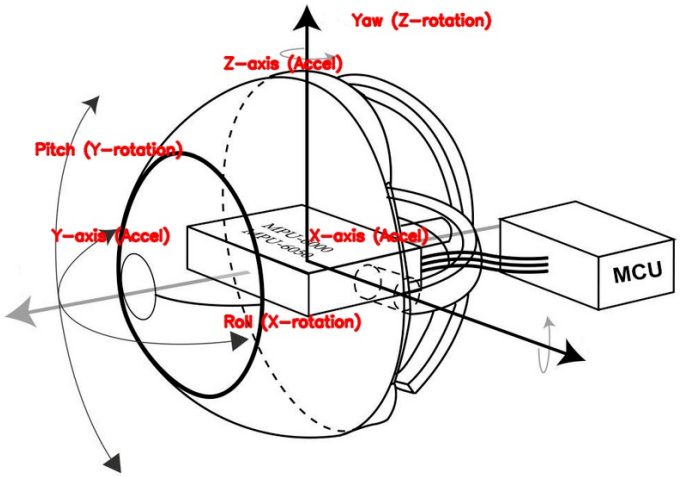


Fig. 4.1: IMU Axes Representation: Accelerometer (Linear Motion) and Gyroscope (Rotational Motion)

4.1.1 Understanding Accelerometers: Measuring Linear Motion

An accelerometer is a sensor that measures linear acceleration along 3 axes, detecting how fast an object is speeding up or slowing down in a given direction. It operates based on different principles, with the most common being the Micro-Electro-Mechanical Systems (MEMS) accelerometer. Inside the sensor, a microscopic suspended mass (proof mass) is attached to springs. When the device moves, the mass shifts due to inertia, causing a change in the distance between capacitor plates, which alters the capacitance. This variation is then converted into an electrical signal, allowing the system to determine acceleration. Accelerometers can measure acceleration along three axes (X, Y, Z) and can detect both static acceleration (such as gravity) and dynamic acceleration (such as movement and vibrations). They are available in different ranges (e.g., $\pm 2g$, $\pm 16g$) to suit various applications, from motion tracking and tilt sensing in smartphones to vibration monitoring in industrial systems and inertial navigation in aerospace and autonomous vehicles.

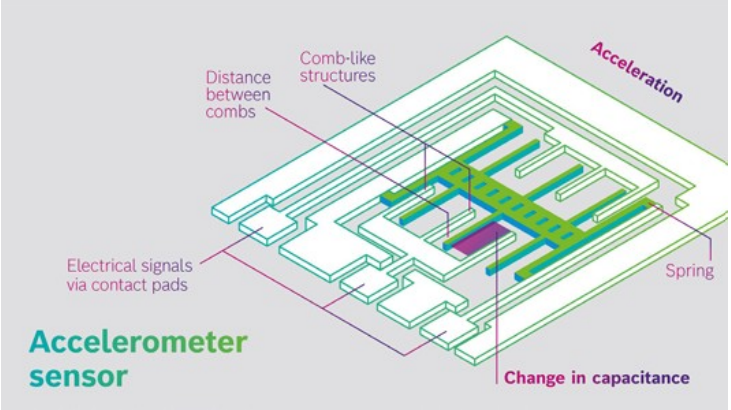


Fig. 4.2: MEMS Accelerometer Structure and Working Principle

4.1.2 Gyroscopes: Tracking Rotational Movement

A gyroscope is a sensor that measures angular velocity or the rate of rotation around one or more axes. It detects how fast an object is rotating and in which direction. The most common type used today is the Micro-Electro-Mechanical Systems (MEMS) gyroscope, which operates based on the Coriolis effect. Inside the sensor, a tiny vibrating mass moves when the device rotates. This movement causes a shift in the vibration pattern due to the Coriolis force, which is detected by capacitive sensing. The change in capacitance is then converted into an electrical signal, which is processed to determine the rate of rotation. Gyroscopes can measure rotation along one, two, or three axes (X, Y, Z) and are used to track orientation changes, stability, and motion control. They are commonly found in inertial navigation systems, drones, robotics, gaming controllers, virtual reality (VR) headsets, and automotive stability control systems. Unlike accelerometers, which detect linear motion, gyroscopes specifically measure rotational motion, making them essential for precise orientation tracking.

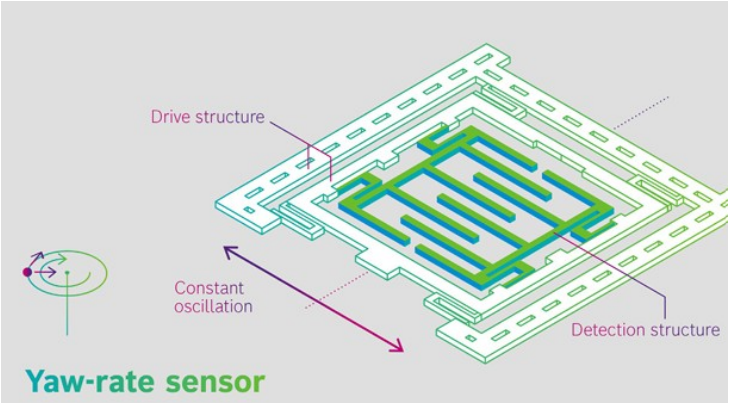


Fig. 4.3: MEMS Gyroscope Structure and Working Principle

4.1.3 Magnetometers: Enhancing Orientation with Earth's Magnetic Field

A magnetometer is a sensor that measures the Earth's magnetic field along one or more axes to determine direction and orientation. It detects the strength and direction of magnetic fields and is commonly used for navigation and heading estimation. The most common type used today is the Micro-Electro-Mechanical Systems (MEMS) magnetometer, which operates based on the Hall effect or magneto-resistive principles. Inside the sensor, a magnetic field interacts with the sensing element, causing a change in electrical resistance or voltage, which is then processed to determine the magnetic field's direction and strength. Magnetometers can measure magnetic field components along three axes (X, Y, Z) and are primarily used as digital compasses to correct orientation drift in IMUs when combined with accelerometers and gyroscopes. They are widely used in navigation systems, smartphones, drones, robotics, and geological studies. Unlike accelerometers and gyroscopes, which measure linear and angular motion, magnetometers specifically detect magnetic field variations, making them crucial for absolute heading determination in motion tracking and positioning application.

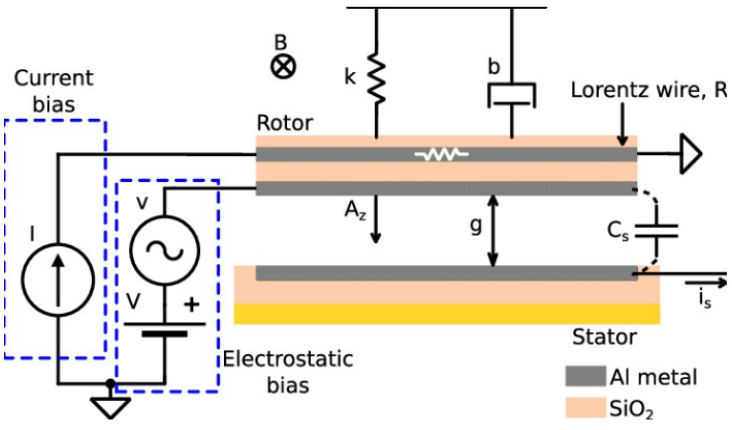


Fig. 4.4: MEMS Magnetometer Structure and Working Principle

4.2 IMU Classification Based on Degrees of Freedom

IMUs (Inertial Measurement Units) are classified based on their Degrees of Freedom (DOF), which refers to the number of independent motion parameters they can measure. The classification is determined by the combination of accelerometers, gyroscopes, and magnetometers used in the IMU. The main types of IMUs based on DOF are:

4.2.1 3-DOF IMU

- **Sensors Included:** Only a 3-axis accelerometer
- **Measures:** Linear acceleration (X, Y, Z)
- **Limitations:** Cannot measure rotation or orientation changes.
- **Applications:** Basic tilt sensing and vibration monitoring.

4.2.2 6-DOF IMU

- **Sensors Included:** 3-axis accelerometer + 3-axis gyroscope
- **Measures:** Linear acceleration and angular velocity (X, Y, Z)
- **Limitations:** Cannot measure absolute heading without external references.
- **Applications:** Drones, robotics, inertial navigation, and motion tracking.

4.2.3 9-DOF IMU

- **Sensors Included:** 3-axis accelerometer + 3-axis gyroscope + 3-axis magnetometer
- **Measures:** Linear acceleration, angular velocity, and Earth's magnetic field (X, Y, Z)
- **Advantages:** The magnetometer provides an absolute reference, helping correct gyroscope drift and improving orientation tracking.
- **Applications:** Virtual reality (VR), augmented reality (AR), robotics, and motion capture.

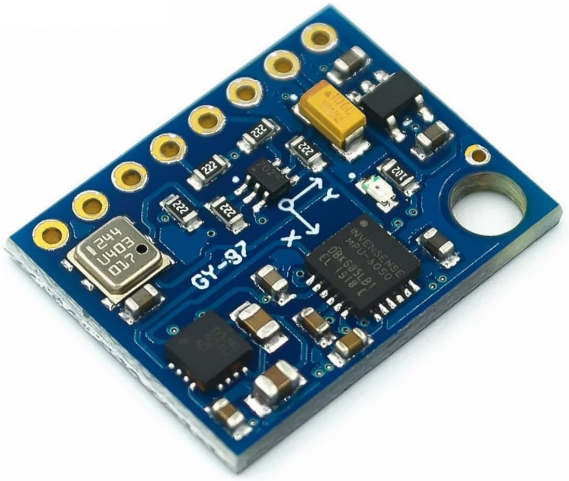
4.3 Choosing an IMU for Hand Orientation Detection

Several commercially available IMUs were researched, offering 6 to 9 degrees of freedom (DoF) with combinations of accelerometers, gyroscopes, and, in some cases, magnetometers. The selection process considered key factors such as sampling frequency, communication interfaces (I²C or SPI), degrees of freedom, and cost-effectiveness.

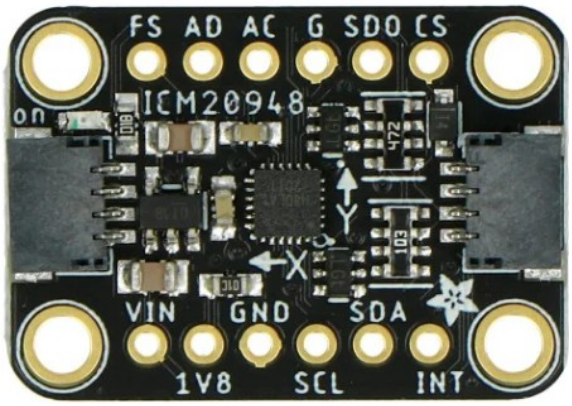
IMU Sensor	Degrees of Freedom (DoF)	Communication Interface
BNO085	9 DoF	I²C/SPI
BMI270	6 DoF	I²C/SPI
BMI160	6 DoF	I²C/SPI
ICM20948	9 DoF	I²C/SPI
MPU6050	6 DoF	I²C
MPU9250	9 DoF	I²C
BNO065	9 DoF	I²C/SPI

Tab. 4.1: Comparison of IMU Sensors Based on DoF and Communication Interface

After careful consideration, one 6-DoF IMU and one 9-DoF IMU were selected for hand movement tracking. The MPU6050 Figure 4.5 (a) is a 6-axis IMU that integrates a 3-axis gyroscope and a 3-axis accelerometer, making it a cost-effective choice for basic motion tracking. On the other hand, the ICM20948 Figure 4.5 (b) is a 9-axis IMU that includes a 3-axis magnetometer (AK09916), providing enhanced orientation tracking. Additionally, the ICM20948 offers lower power consumption, making it more suitable for battery-powered applications. Another advantage of the ICM20948 is its support for both I²C and SPI interfaces, enabling faster communication, whereas the MPU6050 is limited to I²C. Both IMUs were selected for further testing, where different motion and position tracking algorithms were employed to evaluate and compare their accuracy and performance.



(a) MPU6050 IMU



(b) ICM20948 IMU

4.3.1 Limitations and Challenges of IMUs

IMUs (Inertial Measurement Units) are powerful motion-tracking sensors but come with several challenges. One major issue is **gyro drift**, where gyroscopes accumulate small errors over time, causing orientation deviations. IMUs also lack absolute positioning since they only measure changes in motion, leading to increasing errors over time. External factors like mechanical vibrations and shocks can introduce noise, especially in MEMS-based IMUs. For 9-DOF IMUs, magnetic interference from nearby electronics or metal objects can affect accuracy, requiring careful placement and calibration. Another significant problem is **gimbal lock**, which happens when two rotational axes align, causing a loss of one degree of freedom. This makes it difficult to track certain orientations correctly, leading to sudden jumps or incorrect motion data. Additionally, IMUs consume power continuously, and their accuracy can degrade over time due to sensor biases, temperature changes, and aging effects. There is also a tradeoff between cost and performance—high-precision IMUs are expensive, while affordable MEMS-based IMUs suffer from higher noise and drift.

To overcome the disadvantages of IMUs, sensor fusion techniques are employed, which combine data from multiple sensors to enhance accuracy, reduce drift, and provide reliable motion tracking. IMUs suffer from inherent issues such as sensor drift, noise, and lack of absolute positioning, which can degrade their performance over time. Sensor fusion algorithms help mitigate these issues by filtering errors, correcting drift, and improving motion estimation. Without sensor fusion, an IMU alone cannot provide stable and accurate orientation, leading to cumulative errors that make long-term motion tracking unreliable.

4.3.2 Introduction to Sensor Fusion

Sensor fusion is the process of combining data from multiple sensors—typically accelerometers, gyroscopes, and magnetometers—to achieve a more precise and stable representation of an object's movement and orientation. Each of these sensors has unique strengths and weaknesses, and sensor fusion takes advantage of their complementary characteristics to overcome individual limitations.

- **Accelerometers** measure linear acceleration and help detect motion and tilt. However, they are highly susceptible to noise and cannot differentiate between gravitational acceleration and actual movement, making them unreliable for tracking rotation alone.
- **Gyroscopes** measure angular velocity, which allows for precise tracking of rotational movements. However, they suffer from drift, where small errors accumulate over time, leading to significant inaccuracies in orientation estimation.
- **Magnetometers** measure the Earth's magnetic field, providing an absolute heading reference, similar to a digital compass. However, they are susceptible to magnetic interference from nearby electronic devices or metal objects, which can distort readings and affect accuracy.

By integrating and filtering data from these sensors, sensor fusion algorithms effectively eliminate the weaknesses of individual sensors and provide more precise motion tracking. For example, the gyroscope can offer short-term precision in tracking orientation, but its drift is corrected by the accelerometer and magnetometer. The accelerometer can sense gravity and detect static orientation, while the magnetometer ensures an accurate heading direction. Together, these three sensors create a more robust motion tracking system that is less prone to drift, noise, and inaccuracies.Sensor fusion is crucial for applications requiring precise motion tracking, such as drones, robotics, augmented reality (AR), virtual reality (VR), navigation systems, and autonomous vehicles. By implementing sensor fusion techniques, the performance of IMUs is significantly improved, making them more reliable for real-world applications.

4.4 Challenges in Orientation Tracking Without Sensor Fusion

4.4.1 Using Only an Accelerometer

An accelerometer estimates orientation by measuring the direction of gravity relative to the sensor, providing values along the X, Y, and Z axes. By analyzing these acceleration components, roll (tilt around the X-axis) and pitch (tilt around the Y-axis) can be calculated using trigonometric functions:

$$\phi = \arctan\left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}}\right) \tag{4.1}$$

$$\theta = \arctan\left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}}\right) \tag{4.2}$$

However, the accelerometer cannot measure yaw (rotation around the Z-axis) because it detects only linear acceleration, not rotational motion. Additionally, external forces such as vibrations or movements introduce errors, making raw accelerometer-based orientation unreliable. To improve accuracy, accelerometer readings should be combined with a gyroscope for smooth short-term rotation tracking and a magnetometer for absolute heading estimation. Sensor fusion techniques like the Complementary Filter, Madgwick Filter, or Kalman Filter help correct drift and enhance motion tracking, ensuring more accurate and stable orientation estimation in real-world applications.

4.4.2 Using Only a Gyroscope

A gyroscope measures angular velocity around the X, Y, and Z axes and can be used to calculate orientation by integrating these angular velocities over time. Unlike an accelerometer, a gyroscope does not suffer from external acceleration disturbances, making it more reliable for short-term orientation tracking. However, gyroscopes have a major limitation: drift. Over time, small errors accumulate in the integration process, causing the estimated orientation to deviate from the true value.To compute the orientation angles (roll, pitch, and yaw), we integrate the gyroscope's angular velocity readings over time:

$$\phi(t) = \phi(t-1) + \omega_x \cdot \Delta t \tag{4.3}$$

$$\theta(t) = \theta(t-1) + \omega_y \cdot \Delta t \tag{4.4}$$

$$\psi(t) = \psi(t-1) + \omega_z \cdot \Delta t \tag{4.5}$$

where:

- ϕ is the roll (rotation around the X-axis),
- θ is the pitch (rotation around the Y-axis),
- ψ is the yaw (rotation around the Z-axis),
- $\omega_x, \omega_y, \omega_z$ are the angular velocities from the gyroscope along the three axes,
- Δt is the time interval between measurements.

4.4.3 Disadvantages of Both Approaches

Both accelerometer-based and gyroscope-based approaches have significant disadvantages when used independently for orientation estimation:

- **Accelerometer:** Highly susceptible to external forces, vibrations, and movement, making it unreliable in dynamic environments. While it can estimate roll and pitch using gravity, it cannot measure yaw and is affected by high-frequency noise, requiring heavy filtering. **Good for static orientation detection, but unreliable in dynamic conditions.**
- **Gyroscope:** Provides smooth and responsive tracking by integrating angular velocity, but suffers from drift—small errors accumulate over time, causing orientation estimates to deviate significantly. Since it measures only relative changes in rotation and has no absolute reference, it cannot correct long-term errors without external assistance. **Good for dynamic motion tracking, but unreliable for static orientation.**

Due to these limitations, sensor fusion techniques are necessary to combine the strengths of both sensors—using the accelerometer for long-term stability and the gyroscope for short-term precision to achieve accurate and drift-free orientation tracking.

4.5 Solution Using a Complementary Filter

To overcome the disadvantages of using either an accelerometer or a gyroscope alone for orientation estimation, a **Complementary Filter** is commonly used as a simple and efficient sensor fusion technique. This filter combines the strengths of both sensors—using the accelerometer to provide long-term stability and the gyroscope for short-term precision—while minimizing their individual weaknesses. The gyroscope provides smooth and responsive rotation tracking, but its drift accumulates over time. Meanwhile, the accelerometer offers an absolute reference for orientation by detecting gravity but is highly susceptible to external acceleration and noise. The Complementary Filter blends these two sources of data using a weighted approach, where the gyroscope contributes high-frequency data for immediate response, and the accelerometer provides low-frequency corrections to counteract drift.

Mathematically, the filter can be expressed as:

$$\phi_{\text{final}} = \alpha(\phi_{\text{gyro}} + \omega \cdot \Delta t) + (1 - \alpha)\phi_{\text{accel}} \tag{4.6}$$

$$\theta_{\text{final}} = \alpha(\theta_{\text{gyro}} + \omega \cdot \Delta t) + (1 - \alpha)\theta_{\text{accel}} \tag{4.7}$$

where:

- ϕ_{final} and θ_{final} are the filtered roll and pitch angles,
- ϕ_{gyro} and θ_{gyro} are the roll and pitch estimates from the gyroscope, given by:

$$\phi_{\text{gyro}} = \phi_{\text{previous}} + \omega_x \cdot \Delta t \tag{4.8}$$

$$\theta_{\text{gyro}} = \theta_{\text{previous}} + \omega_y \cdot \Delta t \tag{4.9}$$

- ϕ_{accel} and θ_{accel} are the roll and pitch estimates from the accelerometer, calculated as:

$$\phi_{\text{accel}} = \arctan\left(\frac{a_y}{a_z}\right) \tag{4.10}$$

$$\theta_{\text{accel}} = \arctan\left(\frac{-a_x}{\sqrt{a_y^2 + a_z^2}}\right) \tag{4.11}$$

- ω is the angular velocity from the gyroscope,
- Δt is the time interval between updates,
- α is the weighting factor (typically around 0.98 for the gyroscope and 0.02 for the accelerometer).

4.6 Disadvantages of the Complementary Filter

While the Complementary Filter is a simple and efficient solution for sensor fusion, it has several disadvantages. One major limitation is that it does not estimate yaw (rotation around the Z-axis), as it only fuses accelerometer and gyroscope data; a magnetometer is needed for accurate heading correction. Additionally, the filter relies on a fixed weighting factor (α), which may not be optimal in all conditions—a higher α gives more importance to the gyroscope but increases drift, while a lower α reduces drift but makes the system more sensitive to accelerometer noise and external forces. This tradeoff makes the filter less adaptive to varying motion dynamics. Furthermore, it assumes linear sensor behavior and does not account for complex, non-linear errors, which Kalman Filters can handle more effectively. While it provides a reasonable balance between short-term precision and long-term stability, it is not as accurate as more advanced sensor fusion algorithms (such as the Madgwick or Kalman Filters) in highly dynamic environments where sudden accelerations or rotations occur.

4.7 Kalman Filter for Sensor Fusion

To overcome the limitations of the Complementary Filter, a more advanced sensor fusion technique, the Kalman Filter (KF), is used. The Kalman Filter is a predictive algorithm that continuously estimates and corrects orientation by considering uncertainties in sensor measurements. Unlike the Complementary Filter, which relies on a fixed weighting factor, the Kalman Filter dynamically adjusts the weight given to the gyroscope and accelerometer data based on their respective noise levels. This makes it more adaptive and accurate, especially in environments with highly dynamic motion.

4.7.1 Operation of the Kalman Filter

The Kalman Filter operates in two main steps:

1. **Prediction Step:** The filter predicts the current state (orientation) based on the previous state and the gyroscope's angular velocity, updating the estimate over time:

$$x_k = Ax_{k-1} + Bu_k + w_k \tag{4.12}$$

   where x_k is the estimated state (orientation), A is the state transition matrix, B represents the control input (gyroscope readings), and w_k is the process noise.

2. **Correction (Update) Step:** The filter corrects the predicted state using measurements from the accelerometer (and optionally a magnetometer for yaw correction). It applies a Kalman Gain (K) to optimally blend the gyroscope and accelerometer data:

$$x_k = x_k + K(z_k - Hx_k) \tag{4.13}$$

   where z_k is the accelerometer measurement, H is the measurement matrix, and K is the Kalman Gain, which determines the weight assigned to the new measurement versus the prediction.

4.7.2 Advantages and Disadvantages

The Kalman Filter provides several advantages over the Complementary Filter, making it a more robust solution for sensor fusion in IMUs:

- **Dynamic Weighting:** Unlike the Complementary Filter, which uses a fixed blending factor, the Kalman Filter automatically adjusts the influence of each sensor based on real-time noise estimation.
- **Drift Correction:** It continuously refines the orientation estimate, significantly reducing gyroscope drift over time.
- **Robustness in Dynamic Motion:** The filter performs well in high-speed movements, sudden accelerations, and external disturbances, making it ideal for applications such as robotics, drones, and navigation systems.
- **Yaw Estimation:** By incorporating a magnetometer, the Kalman Filter can estimate yaw, a limitation of the Complementary Filter.

However, despite these benefits, the Kalman Filter has some drawbacks:

- **Computational Complexity:** It requires significantly more processing power compared to the Complementary Filter, making it less suitable for low-power embedded systems.
- **Noise Modeling Sensitivity:** The filter's performance depends on accurate noise covariance tuning. Poorly configured parameters can lead to suboptimal results or instability.
- **Does not solve gimbal lock:** The Kalman Filter still suffers from the issue of **gimbal lock**, a fundamental problem in Euler angle-based orientation representation, where the loss of one degree of freedom occurs due to axis alignment.

5 Quaternion-Based Sensor Fusion: Solving Gimbal Lock and Drift

5.1 Introduction

While the Kalman Filter significantly improves sensor fusion by dynamically adjusting weighting and reducing gyroscope drift, it does not solve the fundamental issue of **gimbal lock**. Gimbal lock occurs when two rotation axes align, causing the loss of one degree of freedom in Euler angle-based systems. To overcome this limitation, a more robust mathematical representation is required.Through research, I identified **quaternions** as the ideal solution to the gimbal lock problem. Quaternions offer a four-dimensional representation of orientation, preventing singularities and ensuring smooth rotational calculations. However, the challenge remained in finding a sensor fusion algorithm that outputs quaternions while also mitigating gyroscope drift.Upon further investigation, I discovered a quaternion-based complementary filter. This technique combines the advantages of the complementary filter with quaternion representation, solving gimbal lock while keeping the gyroscope drift in check.

The following sections will explore the fundamentals of quaternions, their application in sensor fusion, and the quaternion-based complementary filter that provides an effective solution for real-time orientation tracking.

Here's the basics of what this Quaternion inspired complementary filter logic has to offer:

1. Quaternion-based orientation estimate.
2. Never requires re-orthogonalization, unlike DCM or any other 3×3 rotation matrix solution.
3. Almost never requires re-normalization of the attitude estimate quaternion.
4. Requires no trigonometric functions or floating-point division, so it runs fast.
5. No gimbal lock in any orientation.
6. Fully 3D capable after initialization. It will properly counter attitude estimate drift and error regardless of the amount of time in any orientation.

5.2 Estimating Orientation Changes via Gyro-Derived Quaternions

Accurately estimating an object's orientation in space is critical for motion tracking applications, such as hand rotation detection. The gyroscope plays a key role in this process by measuring angular velocity along three perpendicular axes. However, integrating gyroscope data over time introduces an issue known as **gyro drift**, where small measurement errors accumulate, leading to incorrect long-term orientation estimates.

To achieve reliable attitude estimation, a mathematical framework is required that:

- Updates the orientation estimate with each new sensor measurement.
- Converts between an object's local frame of reference and the global frame.
- Avoids gimbal lock, a fundamental limitation of Euler angles.

5.2.1 Why Use Quaternions?

Rotation matrices and quaternions are two common approaches for representing orientation. While rotation matrices use a 3×3 structure, they require more computations and are prone to accumulating numerical errors. Quaternions, on the other hand, are represented by four values (one real and three imaginary components) and offer several advantages:

- They do not suffer from orthogonality errors, eliminating the need for frequent corrections.
- They require fewer multiplications and additions, making them computationally efficient.
- They occupy less memory (four values instead of nine for a rotation matrix).

5.2.2 Fast Quaternion Update from Gyroscope Data

To estimate attitude using quaternions, gyroscope readings must be converted into incremental quaternion updates. The standard approach involves computing an angle-axis representation from the gyroscope data and converting it into a quaternion. However, this method requires trigonometric functions and square roots, making it computationally expensive.

A more efficient approach involves using the small-angle approximation, which simplifies the quaternion update equations:

$$t_2 = \frac{\Delta t}{2} \tag{5.1}$$

$$Q_x = \text{gyro}_x \times t_2 \tag{5.2}$$

$$Q_y = \text{gyro}_y \times t_2 \tag{5.3}$$

$$Q_z = \text{gyro}_z \times t_2 \tag{5.4}$$

$$Q_w = 1.0 - 0.5 \times (Q_x^2 + Q_y^2 + Q_z^2) \tag{5.5}$$

The updated quaternion is then obtained by multiplying the new quaternion increment with the previous attitude quaternion:

$$Q_{\text{current}} = Q_{\text{gyro}} \times Q_{\text{last}} \tag{5.6}$$

The multiplication order depends on whether the rotation is measured in the object's local frame or the world frame.

5.2.3 Advantage

By leveraging quaternions and small-angle approximations, significant computational efficiency can be achieved. For instance, on a low-power microcontroller , quaternion-based updates execute faster than traditional rotation matrix updates, freeing up processing power for other real-time tasks. This efficiency makes quaternion-based filters well-suited for applications requiring rapid and accurate motion tracking.

5.3 Gyro Drift Compensation using Accelerometer with 6DoF Sensor Fusion

While quaternions provide an effective means of tracking orientation, relying solely on gyroscope data introduces a significant issue: **gyro drift**. Over time, small errors in gyroscope measurements accumulate, causing the estimated orientation to gradually deviate from the true value. Since gyroscopes measure rotational velocity rather than absolute orientation, an external reference is required to correct this drift. The accelerometer serves this role by providing a long-term reference for pitch and roll estimation.

5.3.1 The Role of the Accelerometer in Drift Compensation

An accelerometer measures linear acceleration along three orthogonal axes. In the absence of external forces, it primarily detects the acceleration due to gravity. This property allows it to act as a natural reference for determining the true direction of "down." By comparing the accelerometer's gravity measurement with the estimated gravity direction derived from the gyroscope, drift in the pitch and roll angles can be identified and corrected. However, accelerometer readings are affected by external accelerations, vibrations, and noise, making direct corrections unreliable. Instead, a gradual correction approach is used, where small adjustments are applied over time to align the estimated orientation with the accelerometer's reference while filtering out transient disturbances.

5.3.2 Drift Correction Process

The process of compensating for drift using an accelerometer consists of several key steps:

Step 1: Transforming Accelerometer Readings into the Earth Frame

Accelerometer data is provided in the IMU's body frame, meaning the measurements are relative to the sensor's current orientation. To make meaningful comparisons, these readings must first be transformed into the global (earth) frame using quaternion rotation:

$$\mathbf{V}_{\text{accel,earth}} = \mathbf{Q}_{\text{orientation}} \otimes \mathbf{V}_{\text{accel,body}} \otimes \mathbf{Q}^*_{\text{orientation}} \tag{5.7}$$

where:

- $\mathbf{V}_{\text{accel,body}}$ represents the accelerometer measurement in the IMU's body frame.
- $\mathbf{Q}_{\text{orientation}}$ is the current estimated orientation quaternion.
- $\mathbf{Q}^*_{\text{orientation}}$ is the conjugate of the orientation quaternion, used to reverse the transformation.
- $\mathbf{V}_{\text{accel,earth}}$ is the transformed accelerometer vector in the global reference frame.

Performing filtering in the global frame rather than the local frame reduces distortions caused by dynamic movements.

Step 2: Identifying Orientation Error

To detect misalignment, the rotated accelerometer reading is compared to the expected gravity vector, $\mathbf{g} = (0, 0, 1)$. The difference between these two vectors reveals the drift-induced error in the orientation estimate. This is calculated using the cross product:

$$\mathbf{V}_{\text{correction}} = \mathbf{V}_{\text{accel,earth}} \times \mathbf{g} \tag{5.8}$$

If the accelerometer's measurement aligns perfectly with gravity, the cross product results in a zero vector, indicating no correction is needed. Otherwise, the resulting correction vector provides the direction and magnitude of the required adjustment.

Step 3: Scaling the Correction Vector

To avoid abrupt corrections that could introduce instability, the correction vector is scaled down before being applied. This prevents overcompensation and ensures smooth convergence of the orientation estimate. A small proportional gain factor α (typically 0.01–0.1) is used:

$$\mathbf{V}_{\text{correction}} = \mathbf{V}_{\text{correction}} \times \alpha \tag{5.9}$$

Step 4: Applying the Correction to the Gyroscope Readings

Rather than modifying the estimated orientation directly, the correction vector is rotated back into the IMU's body frame and incorporated into the next gyroscope update:

$$\mathbf{V}_{\text{gyro,next}} += \mathbf{Q}^{*}_{\text{orientation}} \otimes \mathbf{V}_{\text{correction}} \otimes \mathbf{Q}_{\text{orientation}} \tag{5.10}$$

This approach integrates the correction gradually, preventing sudden jumps in the attitude estimate. Since the gyroscope measures angular velocity, the correction is applied incrementally over multiple iterations, ensuring smooth and stable drift compensation.For instance, in a system with a 400Hz control loop, only 1/400th of the correction is applied per cycle, reducing sensitivity to short-term noise while allowing the orientation estimate to converge over time. The Figure 5.1 gives complete overview of the working of quternion based complimentary filter using 6 DoF.

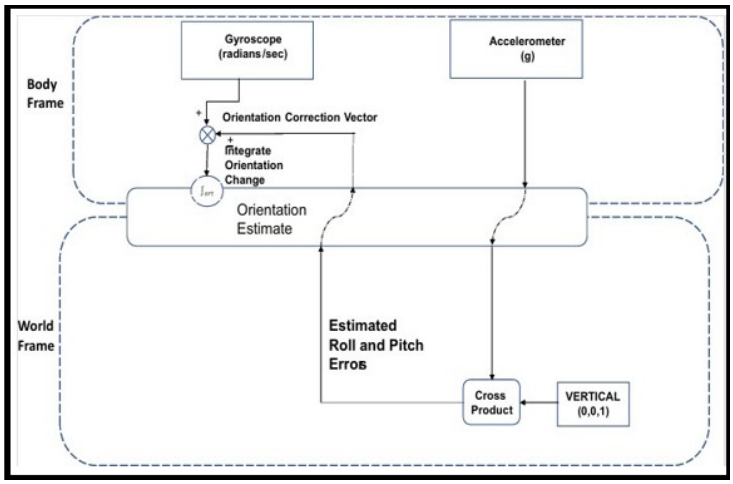


Fig. 5.1: Visual Overview of 6 DoF Quaternion-Based Complementary Filter

Quaternion to Euler Angles Conversion

Given a quaternion representation (q_w, q_x, q_y, q_z), the corresponding Euler angles (Yaw, Pitch, and Roll) can be calculated as follows:

$$\text{Yaw } (\psi) = \text{atan2}\left(2(q_x q_y + q_w q_z), 1 - 2(q_y^2 + q_z^2)\right) \tag{5.11}$$

$$\text{Pitch } (\theta) = \arcsin\left(2(q_x q_z - q_w q_y)\right) \tag{5.12}$$

$$\text{Roll } (\phi) = \text{atan2}\left(2(q_w q_x + q_y q_z), 1 - 2(q_x^2 + q_y^2)\right) \tag{5.13}$$

These formulas allow us to extract Euler angles from a given quaternion while handling the singularities that may occur in the conversion process.

5.3.1 Summary: Using an Accelerometer for Quaternion-Based Gyro Drift Compensation

Gyroscopes alone cannot provide long-term orientation stability due to drift, necessitating external correction sources. Accelerometers serve as a reference for pitch and roll corrections, but their readings are affected by noise and external forces. To address this, drift compensation is performed in four key steps:

1. **Transforming accelerometer data** into the earth frame using quaternion rotation.
2. **Identifying orientation error** by comparing the transformed accelerometer reading with the gravity vector.
3. **Scaling the correction vector** to ensure gradual adjustment and prevent overcorrection.
4. **Applying the correction** to the gyroscope readings before integration, ensuring smooth and stable attitude estimation.

By using this method, gyro drift is continuously corrected without introducing abrupt changes, making it well-suited for high-frequency motion tracking applications. This technique lays the foundation for more advanced sensor fusion approaches, which incorporate additional sensors like magnetometers for full 3D orientation estimation.The next section will explore complementary filter techniques designed to enhance quaternion-based IMU fusion further.

5.4 Magnetometer-Based Yaw Drift Compensation: 9 DoF Sensor fusion

In Inertial Measurement Unit (IMU)-based position and orientation estimation systems, sensor drift is a significant challenge. While gyroscopes provide accurate short-term angular velocity measurements, their readings accumulate errors over time due to integration, leading to drift in orientation estimation. This is particularly problematic for yaw estimation, as there is no inherent correction from accelerometer readings (which only provide tilt information). To mitigate this drift, an additional reference source is required—one that provides absolute orientation data. The magnetometer serves this purpose by measuring the Earth's magnetic field and providing a static reference to magnetic north. This section discusses how magnetometer data is integrated into a complementary filter to correct yaw drift, the challenges associated with using magnetometers, and the techniques employed to improve measurement accuracy.

5.4.1 Magnetometer Functionality

A magnetometer functions similarly to a digital compass. It senses the surrounding magnetic field and provides a 3D vector representing the field's direction in the sensor's frame of reference. By leveraging this information, an IMU system can establish a reference direction and correct yaw drift. The key benefits of using a magnetometer for IMU drift compensation include providing a static reference for absolute yaw estimation. However, integrating magnetometer data is not straightforward due to various real-world distortions and errors that must be accounted for.

5.4.2 Challenges in Using a Magnetometer

Several factors affect the accuracy of magnetometer readings and must be addressed for effective drift compensation.

5.4.3 Hard Iron Effects

- Hard iron distortions are caused by permanent magnetic sources near the sensor, such as ferromagnetic materials or electronic components.
- These effects introduce a constant bias in the magnetometer readings, shifting all measurements in a specific direction.
- The solution involves calibrating the magnetometer by:
  - Identifying the bias in the sensor readings.
  - Subtracting the bias to correct the measurements (hard iron compensation).

5.4.4 Soft Iron Effects

- Soft iron distortions result from nearby materials that distort the magnetic field and vary with orientation.
- When visualized in 3D, affected magnetometer data forms an ellipsoidal shape instead of a perfect circle.
- The solution is to:
  - Apply a 3x3 transformation matrix to reshape the distorted data into a sphere.
  - Perform a thorough calibration to minimize the impact of surrounding ferromagnetic objects.

5.5 Integration into the Complementary Filter

- To effectively use magnetometer data for yaw correction, it must be properly processed and integrated into the sensor fusion algorithm.
- The complementary filter extends the Gyro + Accelerometer system by incorporating the magnetometer.
- This results in a Magnetic, Angular Rate, and Gravity (MARG) system.
- Steps for integration:
  - Transform the magnetometer readings into the world frame.
  - Remove inclination effects by setting the vertical component to zero.
  - Normalize the corrected magnetometer data.
  - Compute yaw correction using the cross product of the estimated north direction and the corrected magnetometer reading.
  - Apply the correction gradually to minimize noise and sudden fluctuations.

5.5.1 Processing Steps for Magnetometer Data

1. **Read Raw Magnetometer Data:** The raw 3D magnetometer readings are collected. Hard iron corrections are applied at this stage.
2. **Transform Data to the World Frame (Tilt Compensation):** Since the magnetometer is mounted on a moving platform, its readings are affected by the device's orientation. The data must be rotated into the world frame using the current orientation estimate.
3. **Remove Inclination:** The vertical component of the magnetic field is eliminated to avoid errors in yaw estimation.
4. **Normalize the Vector:** The magnetometer readings are normalized to create a unit vector, ensuring scale consistency.

5. **Account for Declination:** If true north is required, a correction is applied to adjust for the local magnetic declination.

6. **Compute Yaw Correction using a Cross Product:** The estimated north vector (from the current orientation) is compared with the actual magnetometer readings. The cross product between these two vectors determines the correction vector. If both vectors are identical, no correction is needed. Otherwise, the system applies a correction proportional to the error.

7. **Apply Correction Gradually:** The correction is incrementally applied to the gyro data to avoid sudden changes. This acts as a proportional controller with low gain, ensuring smooth and stable yaw correction.The Figure 5.2 gives complete overview of the working of quternion based complimentary filter using 6 DoF.

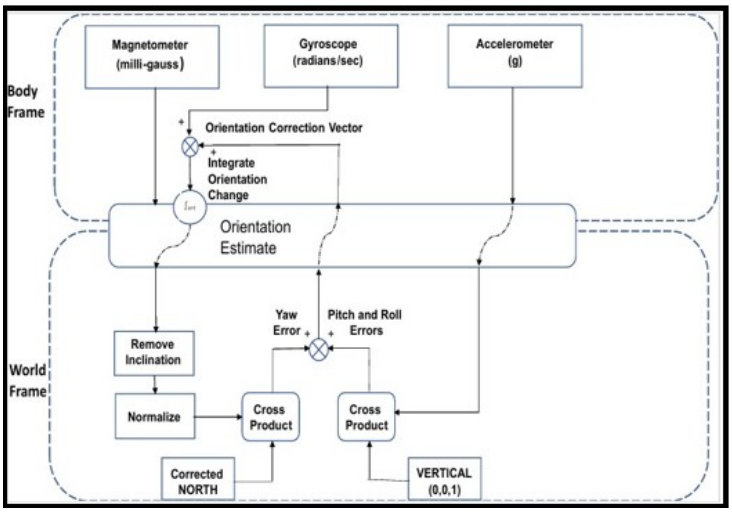


Fig. 5.2: Visual Overview of 9 DoF Quaternion-Based Complementary Filter

5.6 Summary: Magnetometer-Based Yaw Drift Compensation

Using a magnetometer to correct yaw drift significantly enhances the long-term accuracy of IMU-based orientation estimation systems. However, successful implementation requires careful handling of inclination, declination, hard/soft iron effects, and dynamic magnetic interference. By integrating magnetometer data into a complementary filter and applying corrections gradually, it is possible to achieve stable and accurate yaw estimation, enabling reliable navigation and control in real-world applications.

6 Position detection using an IMU

6.1 Position Estimation Using an IMU

Estimating position using an Inertial Measurement Unit (IMU) involves multiple steps, including reading sensor data, compensating for gravity, and integrating acceleration to obtain velocity and position. The following steps outline the process:

6.1.1 Reading IMU Sensor Data

- The raw accelerometer data is collected from the IMU.
- This data includes both linear acceleration and the gravitational component.

6.1.2 Removing Gravity from Accelerometer Data

- The gravity vector is transformed into the IMU's reference frame.
- It is then subtracted from the raw accelerometer reading to obtain the linear acceleration:

$$a_{\text{linear}} = a_{\text{measured}} - g \tag{6.1}$$

  where:
  - a_{linear} is the linear acceleration after removing gravity.
  - a_{measured} is the raw acceleration measured by the IMU.
  - g is the gravity vector, which is typically $(0, 0, 9.81 \text{ m/s}^2)$ in world coordinates.

6.1.3 Integrating Linear Acceleration to Obtain Velocity

- Numerical integration is applied to estimate velocity. A common approach is the trapezoidal rule:

$$v_t = v_{t-1} + a_{\text{linear}} \cdot \Delta t \tag{6.2}$$

  where:
  - v_t is the velocity at time t.
  - v_{t-1} is the velocity at the previous time step.
  - a_{linear} is the linear acceleration computed in the previous step.
  - Δt is the time interval between measurements.
- Small errors in acceleration measurements can accumulate over time, causing velocity drift.

6.1.4 Integrating Velocity to Obtain Position

- A second integration step is performed to determine position:

$$p_t = p_{t-1} + v_t \cdot \Delta t \tag{6.3}$$

  where:

  - p_t is the estimated position at time t.
  - p_{t-1} is the position at the previous time step.
  - v_t is the velocity at time t.
  - Δt is the time interval between measurements.

- This integration is susceptible to drift, requiring additional correction techniques such as sensor fusion with external references (e.g., GPS or visual odometry).

6.2 Limitations and Challenges in IMU-Based Position Estimation

IMU-based position estimation offers a simple and real-time solution for tracking motion. However, due to its reliance on accelerometers and gyroscopes, it suffers from several limitations that affect long-term accuracy and usability. The primary challenges include sensor noise, integration drift, and the lack of an absolute reference frame. IMUs contain accelerometers and gyroscopes that suffer from inherent noise and bias, which impact measurement accuracy. Bias errors in gyroscopes and accelerometers accumulate over time, leading to deviations in velocity and position calculations. Calibration and filtering techniques, such as zero-velocity updates (ZUPTs) and bias estimation, can help mitigate these errors but do not eliminate them completely. IMU-based position estimation relies on integrating acceleration to obtain velocity and then integrating velocity to estimate position. Any small errors in acceleration readings get amplified due to double integration, leading to significant drift over time. Without external corrections, positional drift becomes unbounded, making IMU-based estimation impractical for long-term applications. Additionally, IMUs measure motion relative to their initial position but do not provide absolute position information.

7 Result

7.1 Results and Analysis

To evaluate the performance of the proposed IMU-based tracking system, an experimental setup was designed to imitate the same hand movements as when the glove was worn. The sensor was placed in the same position as it would be on the glove, but instead of the glove, a Vive tracker was used as the reference for comparison, as shown in Figure 7.1. This setup ensured that both the reference (Vive tracker) and the IMU experienced identical movements, allowing for a direct comparison of their outputs.

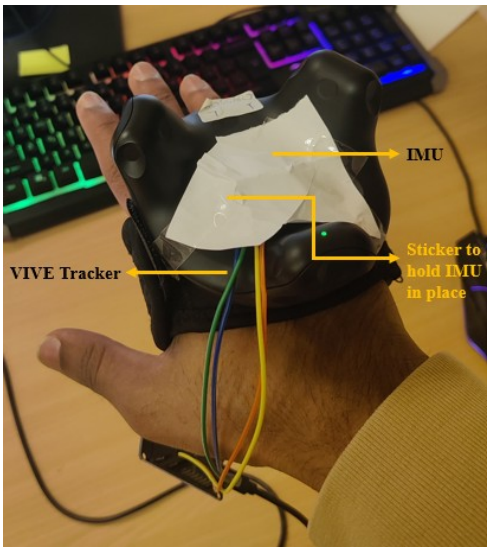


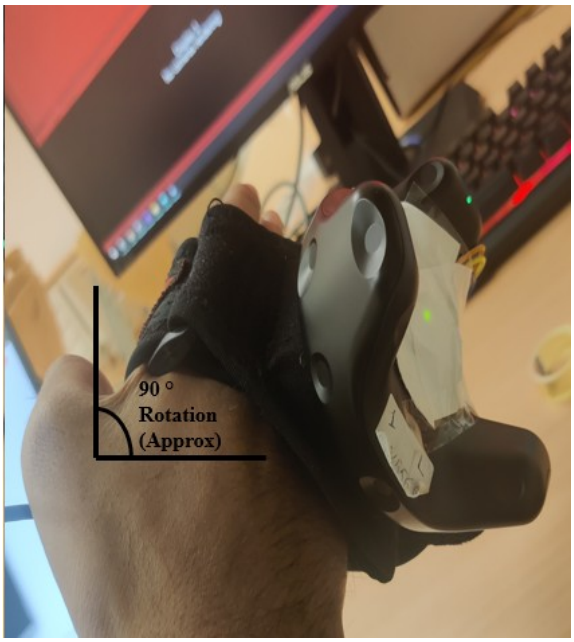
Fig. 7.1: Experimental setup showing the IMU attached to the Vive tracker to replicate the same positioning as on the glove

7.1.1 Orientation Tracking Evaluation

The hand was rotated in controlled motions:

- **90° in one direction** Figure 7.2 (a) and **90° in the opposite direction** Figure 7.2 (b)
- **180° upside down**

For each motion, quaternion values (w, x, y, z) were recorded from both the Vive tracker and the IMU. The experiment was repeated multiple times to ensure consistency. Two different IMUs (ICM20948 and MPU6050) were tested individually under identical conditions.



(a) Approximately 90-Degree Hand Rotation with Vive Tracker



(b) Approximately -90-Degree Hand Rotation with Vive Tracker

Fig. 7.2: Approximately ±90-Degree Hand Rotation with Vive Tracker

7.1.2 Position Tracking Evaluation

To assess the accuracy of position tracking:

- The hand was moved along all three axes ($X,\ Y,\ Z$)
- The positional data ($X,\ Y,\ Z$ coordinates) were recorded for both the Vive tracker and the IMU

These movements were performed multiple times for both IMUs to evaluate the consistency and reliability of their positional tracking.

7.1.3 Data Comparison and Error Analysis

The collected quaternion and position data from both IMUs were compared against the Vive tracker's reference values. The mean absolute error and latency were calculated to quantify the differences.

7.2 Performance Analysis of IMU Sensors

These tables present a performance analysis of two IMU sensors, **ICM20948** and **MPU6050**, in terms of their orientation estimation accuracy and latency. The reference for all measurements is a **Vive Tracker**, which provides reference orientation data. The evaluation is based on three different filtering techniques: **Quaternion Filter, Complementary Filter, and Kalman Filter**.

Sensor	Quaternion Filter (MAE)			Complementary Filter (MAE)			Kalman Filter (MAE)		
	Roll (°)	Pitch (°)	Yaw (°)	Roll (°)	Pitch (°)	Yaw (°)	Roll (°)	Pitch (°)	Yaw (°)
ICM20948	16.12	17.48	29.54	20.42	18.17	35.25	16.17	17.53	29.53
MPU6050	19.55	14.52	79.24	21.50	19.00	88.44	17.05	16.85	77.69

Tab. 7.1: Comparison of Mean Absolute Error (MAE) for ICM20948 and MPU6050 using Quaternion, Complementary, and Kalman Filters.

Sensor	Latency (ms)
ICM20948	20.2
MPU6050	492

Tab. 7.2: Latency comparison for ICM20948, MPU6050, Complementary Filter, and Kalman Filter.

7.2.1 Accuracy Analysis

The Mean Absolute Error (MAE) values for Roll, Pitch, and Yaw angles were analyzed using different filtering techniques. The results indicate that the **Quaternion Filter and the Kalman Filter yield similar performance**, with errors of approximately **16.12° in Roll, 17.48° in Pitch, and 29.54° in Yaw** for the Quaternion Filter, and **16.17° in Roll, 17.53° in Pitch, and 29.53° in Yaw** for the Kalman Filter. The **Complementary Filter exhibits the highest errors**, particularly in Yaw estimation, reaching up to **35.25°**.

7.2.2 ICM20948 vs. MPU6050

- **ICM20948 generally provides lower error values** across all filters, indicating it performs better than MPU6050 in orientation estimation.
- **MPU6050 has particularly high errors in Yaw** (e.g., **79.24° for Quaternion Filter and 88.44° for Complementary Filter**), suggesting that it struggles with heading estimation.

7.2.3 Filter Comparison

- **Quaternion Filter and Kalman Filter yield similar results**, with **Kalman performing slightly better** in most cases.
- **The Complementary Filter generally has the highest errors**, especially for the **Yaw axis**.

7.2.4 Latency Analysis

Latency is another crucial factor in real-time motion tracking applications. The ICM20948 sensor demonstrates a latency of **20.2 ms**, which, while manageable, is notably lower than the **MPU6050's latency of 492 ms**. The high latency of MPU6050 suggests that it may not be suitable for **real-time applications**, whereas ICM20948's lower latency makes it more **responsive and suitable for VR applications**.

7.2.5 ICM20948 vs. Vive Tracker

Since the Vive Tracker serves as the ground truth, the accuracy of ICM20948 was compared against it. The **Yaw angle has the highest error (29-35°)**, meaning ICM20948 struggles more with tracking rotation around the vertical axis. However, **Roll and Pitch errors are lower (16-18°)**, showing that the sensor tracks these axes more accurately.

In terms of latency, the **Vive Tracker's typical latency is 2-5 ms**, whereas **ICM20948 has a latency of 20.2 ms**, introducing an additional delay of about **15-18 ms**. This additional latency might cause **noticeable lag in fast-motion VR applications**, though it remains reasonable for many real-time tasks. The comparison of ICM20948 against MPU6050 and the Vive Tracker highlights some key trade-offs. While **ICM20948 is clearly superior to MPU6050**, offering both **lower error rates and significantly lower latency**, it still falls short when compared to the **Vive Tracker's precision and responsiveness**. The **Yaw error remains the largest drawback**, suggesting that additional sensor fusion techniques—such as **integrating magnetometer corrections or external references**—could improve its performance. Nevertheless, the **low-latency and moderate accuracy** of ICM20948 make it a reasonable choice for applications where **cost and portability** are prioritized over absolute precision.

7.2.6 Why large error?

The large error between the IMU (MPU6050) and the Vive Tracker is primarily due to differences in how they estimate orientation. The Vive Tracker uses external Lighthouse base stations to track its position and orientation in absolute space, meaning it does not drift over time and continuously recalibrates based on external references. In contrast, the IMU relies on internal gyroscopes and accelerometers, which are prone to drift and require filtering algorithms. The filters work by integrating gyroscope data while correcting drift using accelerometer readings. However, when no movement is detected, the filter assumes a steady-state condition and gradually resets the orientation to a neutral or zero-motion state. This causes discrepancies, particularly in yaw, where the IMU drifts and resets while the Vive Tracker remains stable. As a result, when comparing both systems, the IMU shows significant deviation, leading to high MAE values.

7.2.7 Position Comparison

The table 7.3 presents a comparison of the Mean Absolute Error (MAE) in position tracking across three trials using a VIVE Tracker. The MAE values for each trial are reported separately for the x, y, and z coordinates, indicating deviations in position measurements. Trial 1 exhibits relatively consistent errors across all axes, whereas Trial 2 shows a slightly higher error in the x direction. Trial 3, on the other hand, has a notably higher error in the z axis compared to the other trials. These variations suggest minor inconsistencies in position tracking, which could be influenced by sensor noise, environmental factors, or system calibration. Additionally, the observed discrepancies can be attributed to drift accumulation caused by the double integration process over time, which inherently amplifies small errors in acceleration data and leads to increasing deviations in position estimation. The results highlight the accuracy of the VIVE Tracker and potential areas for refinement in position estimation.

Trial	Position x (meters)	Position y (meters)	Position z (meters)
Trial 1	1.0249	1.0240	1.0233
Trial 2	1.0647	1.0268	1.0273
Trial 3	1.0269	1.0294	1.0639

Tab. 7.3: Mean Absolute Error (MAE) comparison of Position calculation across different trials.

8 Conclusion

This project explored the adaptation of an open-source haptic VR glove, originally based on Lucid VR, by integrating IMU-based hand tracking and force feedback mechanisms. By replacing external tracking hardware with IMUs, the system enhances accessibility and portability while maintaining reasonable accuracy.

The evaluation of the IMU-based tracking system highlighted key trade-offs. The ICM20948 outperformed the MPU6050 in both accuracy and latency, making it a viable alternative for applications where cost and mobility are prioritized. However, when compared to the VIVE Tracker, the system exhibited higher yaw error and position drift due to the inherent limitations of IMU-based tracking, particularly the challenges associated with sensor drift and double integration of acceleration data.

While the system does not yet match the accuracy of professional-grade trackers, it presents a promising alternative for applications where cost and portability are key considerations. Future improvements could involve integrating magnetometer corrections, refining sensor fusion techniques, or incorporating external reference points to enhance tracking precision.

By leveraging and refining existing open-source technology, this project contributes to the ongoing efforts to make VR-based medical training more accessible. It underscores the potential of low-cost, open-source solutions in bridging the gap between affordability and functionality in immersive medical simulations.

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